

```

64.     (x, y) tuples of integer indexes for board positions, the values are
65.     WALL, EMPTY_SPACE, or DEAD_ROBOT. The dictionary also has the key
66.     'teleports' for the number of teleports the player has left.
67.     The living robots are stored separately from the board dictionary."""
68.     board = {'teleports': NUM_TELEPORTS}
69.
70.     # Create an empty board:
71.     for x in range(WIDTH):
72.         for y in range(HEIGHT):
73.             board[(x, y)] = EMPTY_SPACE
74.
75.     # Add walls on the edges of the board:
76.     for x in range(WIDTH):
77.         board[(x, 0)] = WALL # Make top wall.
78.         board[(x, HEIGHT - 1)] = WALL # Make bottom wall.
79.     for y in range(HEIGHT):
80.         board[(0, y)] = WALL # Make left wall.
81.         board[(WIDTH - 1, y)] = WALL # Make right wall.
82.
83.     # Add the random walls:
84.     for i in range(NUM_WALLS):
85.         x, y = getRandomEmptySpace(board, [])
86.         board[(x, y)] = WALL
87.
88.     # Add the starting dead robots:
89.     for i in range(NUM_DEAD_ROBOTS):
90.         x, y = getRandomEmptySpace(board, [])
91.         board[(x, y)] = DEAD_ROBOT
92.     return board
93.
94.
95. def getRandomEmptySpace(board, robots):
96.     """Return a (x, y) integer tuple of an empty space on the board."""
97.     while True:
98.         randomX = random.randint(1, WIDTH - 2)
99.         randomY = random.randint(1, HEIGHT - 2)
100.         if isEmpty(randomX, randomY, board, robots):
101.             break
102.     return (randomX, randomY)
103.
104.
105. def isEmpty(x, y, board, robots):
106.     """Return True if the (x, y) is empty on the board and there's also
107.     no robot there."""
108.     return board[(x, y)] == EMPTY_SPACE and (x, y) not in robots
109.
110.
111. def addRobots(board):
112.     """Add NUM_ROBOTS number of robots to empty spaces on the board and
113.     return a list of these (x, y) spaces where robots are now located."""
114.     robots = []
115.     for i in range(NUM_ROBOTS):
116.         x, y = getRandomEmptySpace(board, robots)
117.         robots.append((x, y))
118.     return robots

```